



Circuit Documentation



SUMMARY

This circuit is designed to monitor environmental conditions and control a water pump based on soil moisture levels. It utilizes an Arduino UNO as the central microcontroller to interface with various sensors and modules, including a soil moisture sensor, light sensor (BH1750), temperature and pressure sensor (BME BMP280), and an LTE module for communication. The circuit also includes a relay to control a water pump and an LED indicator.



COMPONENT LIST

1. Arduino UNO

- **Description:** A microcontroller board based on the ATmega328P.
- **Pins:** UNUSED, IOREF, Reset, 3.3V, 5V, GND, Vin, A0, A1, A2, A3, A4, A5, SCL, SDA, AREF, D13, D12, D11, D10, D9, D8, D7, D6, D5, D4, D3, D2, D1, D0.

2. Soil Moisture V2

- **Description:** A sensor to measure soil moisture levels.
- **Pins:** GND, VCC, OUT.

3. a7670E LTE Module

- **Description:** A module for LTE communication.
- **Pins:** GROUND, RX, TX, K, VCC, S.

4. 5V Relay

- **Description:** A relay module to control high-power devices.
- **Pins:** NOOpen, Common, NClosed, VCC, -, S.

5. Resistor (330 Ohm)

- **Description:** A resistor with a resistance of 330 Ohms.
- **Pins:** pin1, pin2.

6. Regulator

- **Description:** A voltage regulator to provide stable power.
- **Pins:** Vin+, Vin-, Vout+, Vout-.



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7. Electrolytic Capacitor (470nF)

- **Description:** A capacitor with a capacitance of 470nF.
- **Pins:** -, +.

8. Ceramic Capacitor (100nF)

- **Description:** A capacitor with a capacitance of 100nF.
- **Pins:** pin0, pin1.

9. Resistor (1k Ohm)

- **Description:** A resistor with a resistance of 1000 Ohms.
- **Pins:** pin1, pin2.

10. Resistor (2k Ohm)

- **Description:** A resistor with a resistance of 2000 Ohms.
- **Pins:** pin1, pin2.

11. BH1750

- **Description:** A light sensor for measuring ambient light.
- **Pins:** ADDR, SDA, SCL, GND, VCC.

12. BME BMP280

- **Description:** A sensor for measuring temperature and pressure.
- **Pins:** VIN, GND, SCL, SDA.

13. 4G Antenna

- **Description:** An antenna for LTE communication.
- **Pins:** Connection.

14. Red LED

- **Description:** An LED for visual indication.
- **Pins:** +, -.

15. 12V Battery (mini)

- **Description:** A power source for the circuit.
- **Pins:** -, +.



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16. Water Pump

- **Description:** A pump for water circulation.
- **Pins:** VCC, GND.

17. Electrolytic Capacitor (2200 μ F)

- **Description:** A capacitor with a capacitance of 2200 μ F.
- **Pins:** -, +.

18. Diode

- **Description:** A diode for current direction control.
- **Pins:** cathode, anode.



WIRING DETAILS

► Arduino UNO

- **GND:** Connected to the common ground net.
- **5V:** Connected to the VCC net for powering sensors and modules.
- **A0:** Connected to the OUT pin of the Soil Moisture V2.
- **A4:** Connected to the SDA pins of BH1750 and BME BMP280.
- **A5:** Connected to the SCL pins of BH1750 and BME BMP280.
- **D10:** Connected to the TX pin of the a7670E LTE Module.
- **D11:** Connected to pin1 of the 1k Ohm Resistor.
- **D7:** Connected to the S pin of the 5V Relay.

► Soil Moisture V2

- **GND:** Connected to the common ground net.
- **VCC:** Connected to the VCC net.
- **OUT:** Connected to A0 on the Arduino UNO.

► a7670E LTE Module

- **GROUND:** Connected to the common ground net.
- **RX:** Connected to pin2 of the 2k Ohm Resistor.
- **TX:** Connected to D10 on the Arduino UNO.
- **VCC:** Connected to the VCC net.
- **S:** Connected to the Connection pin of the 4G Antenna.

► 5V Relay

- **Common:** Connected to the VCC net.
- **VCC:** Connected to the VCC net.
- **-:** Connected to the common ground net.
- **S:** Connected to D7 on the Arduino UNO.
- **NOOpen:** Connected to the + pin of the 2200 μ F Electrolytic Capacitor.



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► Resistor (330 Ohm)

- **pin1:** Connected to NOpen of the 5V Relay.
- **pin2:** Connected to the + pin of the Red LED.

► Regulator

- **Vin-:** Connected to the - pin of the 12V Battery.
- **Vin+:** Connected to the + pin of the 12V Battery.
- **Vout+:** Connected to the VCC net.
- **Vout-:** Connected to the common ground net.

► Electrolytic Capacitor (470nF)

- **-:** Connected to the common ground net.
- **+:** Connected to the VCC net.

► Ceramic Capacitor (100nF)

- **pin0:** Connected to the VCC net.
- **pin1:** Connected to the common ground net.

► Resistor (1k Ohm)

- **pin1:** Connected to D11 on the Arduino UNO.
- **pin2:** Connected to pin2 of the 2k Ohm Resistor.

► Resistor (2k Ohm)

- **pin1:** Connected to pin1 of the 330 Ohm Resistor.
- **pin2:** Connected to RX of the a7670E LTE Module.

► BH1750

- **GND:** Connected to the common ground net.
- **VCC:** Connected to the VCC net.
- **SDA:** Connected to A4 on the Arduino UNO.
- **SCL:** Connected to A5 on the Arduino UNO.

► BME BMP280

- **GND:** Connected to the common ground net.
- **VIN:** Connected to the VCC net.
- **SDA:** Connected to A4 on the Arduino UNO.
- **SCL:** Connected to A5 on the Arduino UNO.



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► 4G Antenna

- **Connection:** Connected to the S pin of the a7670E LTE Module.

► Red LED

- **+**: Connected to pin2 of the 330 Ohm Resistor.
- **-**: Connected to the common ground net.

► 12V Battery (mini)

- **-**: Connected to Vin- of the Regulator.
- **+**: Connected to Vin+ of the Regulator.

► Water Pump

- **VCC:** Connected to NOpen of the 5V Relay.
- **GND:** Connected to the common ground net.

► Electrolytic Capacitor (2200 μ F)

- **-**: Connected to the common ground net.
- **+**: Connected to Diode anode.

► Diode

- **cathode:** Connected to the common ground net.
- **anode:** Connected to Electrolytic Capacitor (2200 μ F) +.